Macroeconomic Shocks and the Transmission of Long-Term Sovereign Bond Yields

A VAR, SVAR, and Cointegration Analysis of U.S. and German 10-Year Government Bond Yields

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Abstract

This paper examines how shocks to a set of macro-financial explanatory variables — industrial production, consumer prices, the policy rate, the central bank balance sheet, and global risk sentiment (VIX) — are transmitted to the U.S. 10-year Treasury yield and the German 10-year Bund yield. The standard multivariate time-series toolkit is applied in sequence: unit-root and stationarity testing, reduced-form VAR estimation with information-criterion lag selection, pairwise Granger causality tests, Johansen cointegration testing and VECM estimation, structural identification via a block-recursive Cholesky decomposition, structural impulse response functions (IRFs), and forecast error variance decomposition (FEVD). The system is ordered as an eleven-variable block-recursive VAR: a six-variable U.S. block, treated as contemporaneously exogenous to a five-variable German/ECB block, following the "global financial cycle" identification logic of Rey (2013) and Miranda-Agrippino and Rey (2020). U.S./global shocks are found to explain roughly two-thirds of the German 10-year yield's forecast error variance at a two-year horizon, versus roughly one-third for domestic ECB shocks; U.S. CPI and VIX shocks are found to Granger-cause the U.S. yield, but the two yields are not found to Granger-cause each other directly; and a single cointegrating relationship is found linking the two yields to the two policy

*iliasbalatsouras@gmail.com. Independent research working paper, prepared following the standard multivariate time-series course sequence (VAR $\rightarrow$ Granger causality $\rightarrow$ cointegration/VECM $\rightarrow$ structural (Cholesky) identification $\rightarrow$ impulse responses $\rightarrow$ variance decomposition). Not peer reviewed; all errors are my own.

rates and the two central bank balance sheets, consistent with a long-run transatlantic monetary-financial equilibrium.

1. Introduction

Government bond yields are among the most closely watched prices in macro-finance: they price the risk-free (or near risk-free) rate, embed market expectations of future monetary policy, and transmit shocks across countries through the actions of globally active investors. A narrow, well-defined question is asked here: given a small system of observable macro-financial variables, what fraction of the U.S. and German 10-year yield's movements can be attributed to domestic monetary policy versus foreign/global shocks, and through what channel — direct Granger-causal predictability, a long-run cointegrating equilibrium, or purely contemporaneous (same-period) transmission — does that attribution operate?

The paper is organized to mirror the standard sequence for analyzing a multivariate, possibly non-stationary system:

1. **Stationarity testing** (Section 3): Augmented Dickey–Fuller (ADF) and KPSS tests are used to establish the order of integration of each series, motivating the choice between a VAR in levels, a VAR in differences, or a vector error-correction model (VECM).
2. **Reduced-form VAR** (Section 4): lag order is selected by information criteria (Sims, 1980), followed by estimation and residual diagnostics.
3. **Granger causality** (Section 5): pairwise predictive-causality tests are run among the policy rates, VIX, and the two yields.
4. **Cointegration and VECM** (Section 6): the Johansen trace and maximum-eigenvalue tests (Johansen, 1991) are applied to the theoretically cointegrated core (yields, policy rates, balance sheets), followed by VECM estimation of the long-run cointegrating vector(s) and short-run adjustment speeds.
5. **Structural (Cholesky) identification** (Section 7): a block-recursive Cholesky decomposition (Sims, 1980; Christiano, Eichenbaum, and Evans, 1999; Bernanke and Blinder, 1992) is applied to the reduced-form VAR residual covariance, with the U.S. block ordered ahead of the German/ECB block.
6. **Structural impulse responses and FEVD** (Section 8): the dynamic and variance-decomposition consequences of the identification in Section 7 are reported.

The paper is placed in the literature in Section 2. Results and limitations are discussed in Section 9; conclusions are drawn in Section 10. Supplementary figures are collected in the Appendix.

2. Literature

2.1. Structural VAR methodology

The recursive (Cholesky) identification scheme used here follows Sims (1980), who first proposed VARs as an atheoretical alternative to large-scale simultaneous-equation macro models, and the recursive-monetary-VAR convention formalized by Bernanke and Blinder (1992) and Christiano, Eichenbaum, and Evans (1999): variables are ordered slow-moving (real activity, prices) $\rightarrow$ policy instrument $\rightarrow$ fast-moving, market-priced (yields, risk sentiment), so that the policy instrument can react contemporaneously to real/price information but asset prices can react contemporaneously to everything. A related but distinct residual-restriction approach is applied by Badinger and Schiman (2023) to identify ECB monetary policy shocks, using high-frequency shock dates around policy announcements as an external identification device – a natural robustness alternative to the recursive scheme used in this paper.

2.2. Global financial cycle and cross-border yield spillovers

A single "global financial cycle," driven substantially by U.S. monetary policy and risk sentiment (proxied here by VIX), is argued by Rey (2013) to co-move asset prices, credit growth, and leverage worldwide regardless of the exchange rate regime. This is formalized by Miranda-Agrippino and Rey (2020) with a global factor extracted from risky asset prices, shown to respond strongly to U.S. monetary shocks and to predict worldwide credit and capital flow cycles. The block-recursive ordering used in this paper – U.S. variables treated as exogenous to the German/ECB block within the period – operationalizes this "small open financial area" assumption for the Euro Area vis-à-vis U.S. Treasury and dollar funding markets.

2.3. Cointegration and error correction

It is established by the Engle–Granger representation theorem (Engle and Granger, 1987) that a set of $I(1)$ variables linked by a long-run equilibrium relationship can be represented as a VECM, separating short-run dynamics from long-run adjustment. The maximum-likelihood, multivariate generalization of this idea (the trace and max-eigenvalue tests used in Section 6) is provided by Johansen (1991). Applied to sovereign yields, a cointegrating relationship between the U.S. and German 10-year yields alongside the Fed funds rate and ECB deposit rate is consistent with a long-run Fisher-type / uncovered-interest-parity-adjacent equilibrium linking policy stances and term premia across the two currency areas – a direction also supported by the ARDL-based evidence in the Levy Institute's work on Eurozone bond yield dynamics (Levy Institute, 2024).

2.4. ECB transmission and core-periphery differentiation

A broader literature on ECB monetary transmission finds that policy pass-through and sovereign risk premia differ substantially between core and peripheral Euro Area sovereigns (ECB, 2013): core issuers such as Germany are found to show a much cleaner, more direct link between the domestic policy rate and the domestic long-term yield than periphery issuers, whose yields carry a larger and more time-varying idiosyncratic credit-risk component. Germany is used here as the "core" Euro Area sovereign deliberately: it is the country least contaminated by idiosyncratic sovereign credit risk, making it the cleanest testing ground for a pure monetary/global-cycle transmission channel to the long end of the yield curve.

3. Data

3.1. Variables

The system comprises eleven monthly series from FRED, spanning January 2003 to March 2024 (255 monthly observations) , block-ordered as follows.

U.S. block (ordered first; slow $\rightarrow$ policy $\rightarrow$ fast): industrial production (`INDPRO`), CPI (`CPIAUCSL`), the federal funds rate (`FEDFUNDS`), the Fed balance sheet in logs (`WALCL`), the VIX (`VIXCLS`), and the 10-year Treasury yield (`GS10`).

Germany/ECB block (ordered second): German industrial production (`DEUPROINDMISMEI`), German HICP (`CP0000DEM086NEST`), the ECB deposit facility rate (`ECBDFR`), the ECB balance sheet in logs (`ECBASSETSW`), and the German 10-year Bund yield (`IRLTLT01DEM156N`).

The two headline yields are plotted over the sample period in Figure 1. Both series show the broad secular decline in long-term interest rates through the 2010s, the post-pandemic reflation and subsequent tightening cycle, and a persistent, time-varying spread between the two countries.

3.2. Stationarity

ADF and KPSS tests on levels and first differences are reported in Table 1. Consistent with standard findings for interest rates, prices, and balance-sheet series, a unit root cannot be rejected in levels for most variables (ADF $p > 0.05$), but is rejected after first differencing, i.e. the series behave as $I(1)$. CPI and the VIX are the partial exceptions: their level ADF/KPSS readings are more ambiguous, consistent with CPI's well-known near-unit-root/trend behavior and the VIX's fast mean reversion. Given the preponderance of $I(1)$ behavior, the reduced-form VAR in Sections 4–5 is estimated in first differences, while cointegration is tested directly in levels in Section 6.

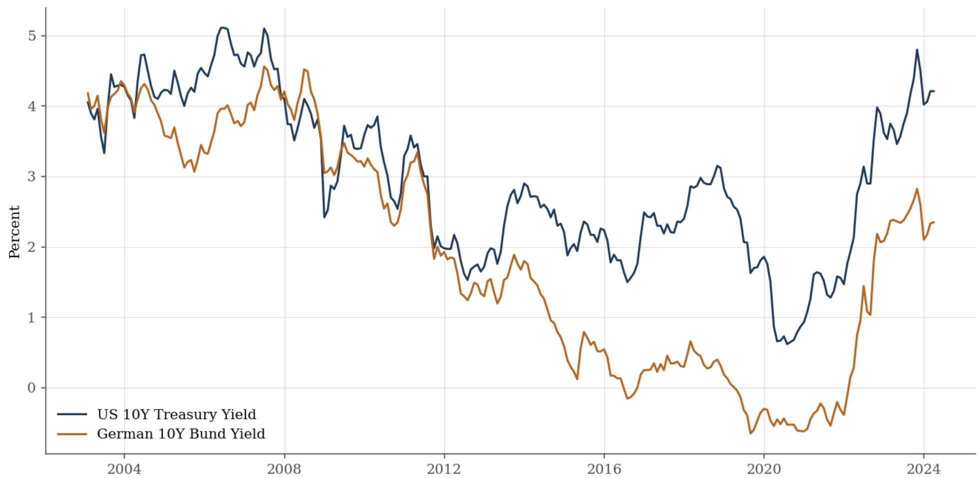

Figure 1: U.S. 10-year Treasury yield and German 10-year Bund yield, levels, full sample.

Table 1: Stationarity tests (selected variables)

Variable	ADF p (level)	KPSS p (level)	ADF p (diff)	Likely $I(1)$
us_10y	0.470	0.010	0.000	Yes
de_10y	0.526	0.010	0.000	Yes
fed_funds	0.124	0.100	0.007	Yes
ecb_rate	0.341	0.024	0.000	Yes
us_cpi	0.999	0.010	0.164	No
de_cpi	0.993	0.010	0.100	No
vix	0.001	0.100	0.000	No
fed_bs	0.691	0.010	0.000	Yes
ecb_bs	0.665	0.010	0.002	Yes

4. Reduced-form VAR

Lag order is selected by AIC, BIC, and HQIC over a maximum of 8 lags; BIC/HQIC favor a single lag, while AIC favors 3. Given that the system has 11 variables and roughly 250 monthly observations, a longer lag would overfit rapidly (each additional lag adds 121 parameters); the VAR is therefore estimated at a deliberately conservative fixed lag of 2. A Portmanteau residual-whiteness test rejects the null of white-noise residuals at the 5% level – a known limitation discussed in Section 9, most likely reflecting volatility clustering in the financial series rather than a fixable lag-order problem.

5. Granger causality

Pairwise Granger causality tests (VAR in differences, 2 lags) are reported in Table 2 for the pairs of primary economic interest.

Table 2: Granger causality tests, selected pairs

Cause	Effect	p-value	Significant (5%)
us_10y	de_10y	0.8929	No
de_10y	us_10y	0.1762	No
fed_funds	us_10y	0.0610	No
ecb_rate	de_10y	0.7976	No
fed_funds	de_10y	0.1169	No
ecb_rate	us_10y	0.6407	No
vix	us_10y	0.0031	Yes
vix	de_10y	0.0528	No
us_cpi	us_10y	0.0001	Yes
de_cpi	de_10y	0.0019	Yes
fed_bs	de_10y	0.0601	No
ecb_bs	de_10y	0.0692	No

Three results stand out. First, the two 10-year yields are *not* found to Granger-cause one another ($p = 0.89$ U.S.$\rightarrow$DE; $p = 0.18$ DE$\rightarrow$US): whatever cross-border transmission exists (shown to be large in Section 8) is not lag-predictable from one yield to the other – it appears to operate contemporaneously or through a third variable, not through the yields' own lagged values. Second, the U.S. yield is found to be Granger-caused by VIX ($p = 0.003$) but only marginally by the German yield ($p = 0.053$), consistent with VIX functioning as a fast-moving, U.S.-anchored global risk indicator. Third, each country's own CPI is found to Granger-cause its own yield (U.S. $p < 0.001$; Germany $p = 0.002$), a direct, lag-predictable inflation pass-through to the long end of the curve, while neither policy rate is found to Granger-cause either yield at conventional significance – policy rate information appears to be absorbed into yields largely contemporaneously (consistent with the efficient-markets expectations-hypothesis view of the term structure) rather than with a predictable lag.

6. Cointegration and error correction

Cointegration is tested among the theoretically linked core of the system – the two 10-year yields, the two policy rates, and the two central bank balance sheets (six variables, in levels) – using the Johansen trace and maximum-eigenvalue tests (Table 3).

Table 3: Johansen cointegration test

H_0: rank $\leq$	Trace stat.	5% crit.	Reject?	Max-eig stat.	5% crit.
0	104.00	95.75	Yes	49.62	40.08
1	54.38	69.82	No	23.67	33.88
2	30.71	47.85	No	12.06	27.59
3	18.65	29.80	No	10.83	21.13
4	7.81	15.49	No	5.57	14.26
5	2.24	3.84	No	2.24	3.84

The null of no cointegration ($r = 0$) is rejected by both tests, while the null of $r \leq 1$ is not rejected, indicating a single cointegrating relationship. Estimating a VECM with this rank yields the cointegrating vector reported in Table 4 (normalized on the U.S. yield).

Table 4: VECM cointegrating vector (normalized on `us_10y`)

Variable	Coefficient
us_10y	1.0000
de_10y	-3.3582
fed_funds	-2.1633
ecb_rate	3.7916
fed_bs	-5.1444
ecb_bs	1.3306

The long-run relationship links the U.S. yield positively to the German yield and the ECB rate, and negatively to the Fed funds rate and the Fed balance sheet – broadly consistent with a Fisher-type / relative-monetary-stance equilibrium (tighter Fed policy relative to the ECB associated with a wider long-run U.S. yield level relative to the German yield). Adjustment speeds (α) are small in magnitude for both yields (U.S. 0.0018, Germany -0.0026 per month), i.e. neither yield is found to error-correct strongly and rapidly back toward the estimated long-run relationship on its own.

7. Structural (Cholesky) identification

The reduced-form VAR residual covariance matrix $\hat{\Sigma}_u$ is factored as $\hat{\Sigma}_u = PP'$ via the Cholesky decomposition, with P lower-triangular under the block-recursive ordering of Section 3: the U.S. block is ordered first (treated as contemporaneously exogenous to Germany/ECB), and within each block, variables are ordered slow $\rightarrow$ policy $\rightarrow$ fast. A one-standard-deviation structural shock to each variable is thereby identified as one that is orthogonal to, and does not contemporaneously respond to, shocks in variables ordered after it.

8. Structural impulse responses and variance decomposition

8.1. Impulse responses

The orthogonalized (structural) impulse response of each yield to shocks in the policy rates, VIX, CPI, and balance sheets, at a 24-month horizon, is plotted separately for the U.S. yield (Figure 2) and the German yield (Figure 3). The response of each yield to the full set of eleven shocks is reported in the Appendix (Figures 6 and 7).

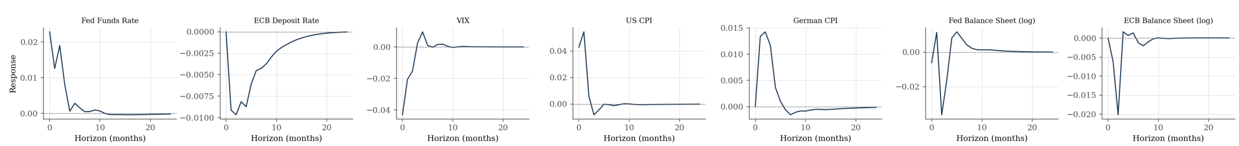

Figure 2: Structural (Cholesky) impulse responses of the U.S. 10-year Treasury yield to key shocks.

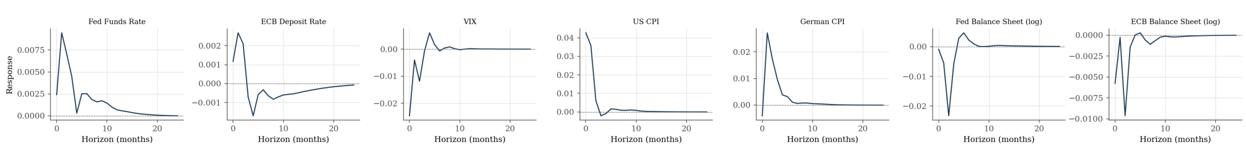

Figure 3: Structural (Cholesky) impulse responses of the German 10-year Bund yield to key shocks.

Consistent with the Granger causality results, a VIX shock and a U.S. CPI shock are found to produce visibly larger and more persistent responses in the U.S. yield than in the German yield, while both policy-rate shocks produce comparatively muted yield responses under this identification – most of the policy-rate-to-yield transmission is captured contemporaneously by the Cholesky ordering itself (the yield is ordered last within each block) rather than showing up as a distinct dynamic IRF path.

8.2. Forecast error variance decomposition

The German 10-year yield's forecast error variance is decomposed in Figure 4 into the cumulative contribution of all six U.S.-block shocks versus all five Germany/ECB-block shocks, at each horizon out to 24 months.

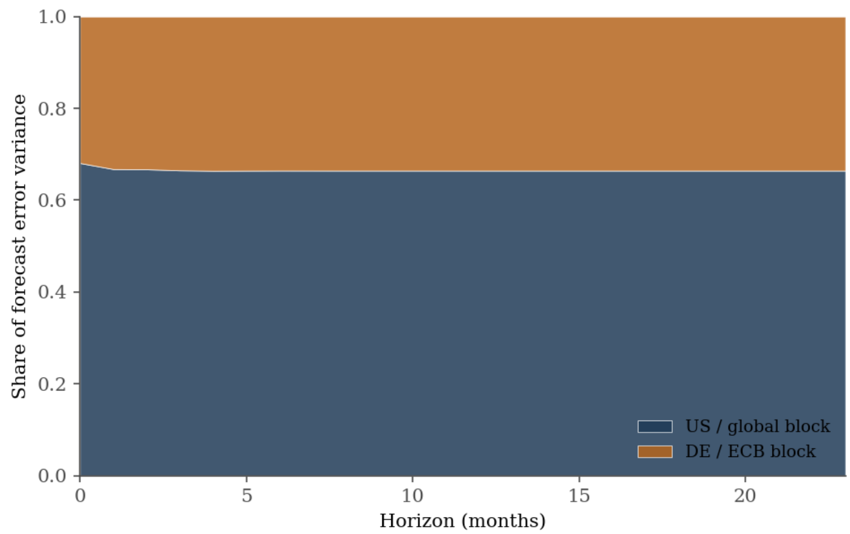

Figure 4: FEVD of the German 10-year Bund yield: cumulative U.S./global block share vs. domestic ECB block share.

At the 24-month horizon, U.S./global shocks are found to account for **66.4%** of the German yield's forecast error variance, versus **33.6%** for domestic ECB-block shocks. This is the headline result of the paper and is consistent with the broader global-financial-cycle literature (Rey, 2013; Miranda-Agrippino and Rey, 2020): even for Germany, the Euro Area's benchmark "risk-free" sovereign issuer, domestic ECB policy is found to explain only a minority of long-term government bond yield variance. By construction of the block-recursive ordering, the U.S. 10-year yield's own forecast error variance is overwhelmingly (94.6% at 24 months) explained by U.S.-block shocks, since the German/ECB block cannot contemporaneously affect the U.S. block under this identification.

9. Discussion and limitations

Three results, taken together, tell a consistent story: German 10-year yields are found to be substantially exposed to U.S./global shocks (FEVD); the channel is not found to be lag-predictable through the U.S. yield's own past values (Granger causality); and the two yields are nonetheless found to share a long-run equilibrium alongside the two policy rates and two balance sheets (cointegration). The most natural reading is that transmission from U.S. to German long-term yields is largely *contemporaneous* – consistent with same-day repricing of Euro Area bonds on U.S. data surprises or Fed announcements, which is correctly attributed by the recursive Cholesky ordering to the U.S. block's structural shocks even though it would not show up as a lagged Granger-causal relationship.

Two limitations should be kept in mind. First, the Portmanteau residual-whiteness test is rejected at every lag specification tried, most plausibly reflecting GARCH-type volatility clustering in the financial variables (VIX, both yields) rather than a mis-specified lag order or cointegration rank; point estimates of sign and rough magnitude should be trusted more than exact percentages. Second, the Cholesky/recursive identification is a strong assumption – results are conditional on the chosen ordering, and the residual-restriction alternative of Badinger and Schiman (2023) discussed in Section 2 suggests a natural robustness check for future work: the same system could be re-identified using high-frequency, event-based shock restrictions rather than a fixed recursive ordering.

10. Conclusion

The standard VAR $\rightarrow$ Granger causality $\rightarrow$ cointegration/VECM $\rightarrow$ Cholesky-SVAR $\rightarrow$ IRF/FEVD sequence is applied to an eleven-variable, block-recursive U.S./Germany-ECB system. It is found that (i) U.S. and German 10-year yields do not Granger-cause each other directly, but (ii) are linked by a single long-run cointegrating relationship together with the two policy rates and two central bank balance sheets, and (iii) U.S./global shocks explain roughly two-thirds of the German yield's forecast error variance at a two-year horizon under a Cholesky identification that treats the U.S. block as contemporaneously exogenous. These results are consistent with the broader "global financial cycle" literature's view that even core Euro Area sovereign yields are only partially a domestic-policy object.

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A. Supplementary figures

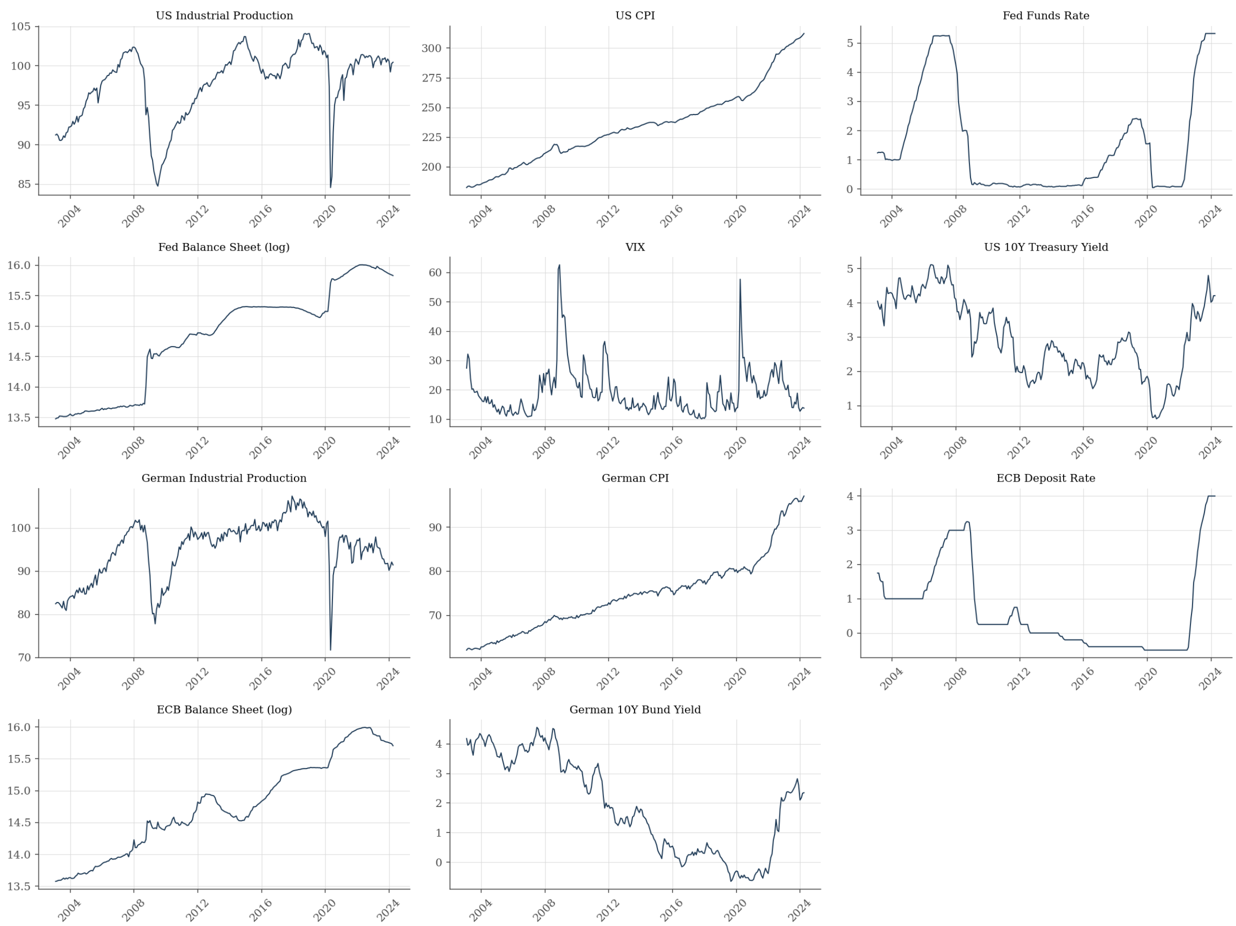

Figure 5: Raw series (levels), all eleven variables in the block-recursive system.

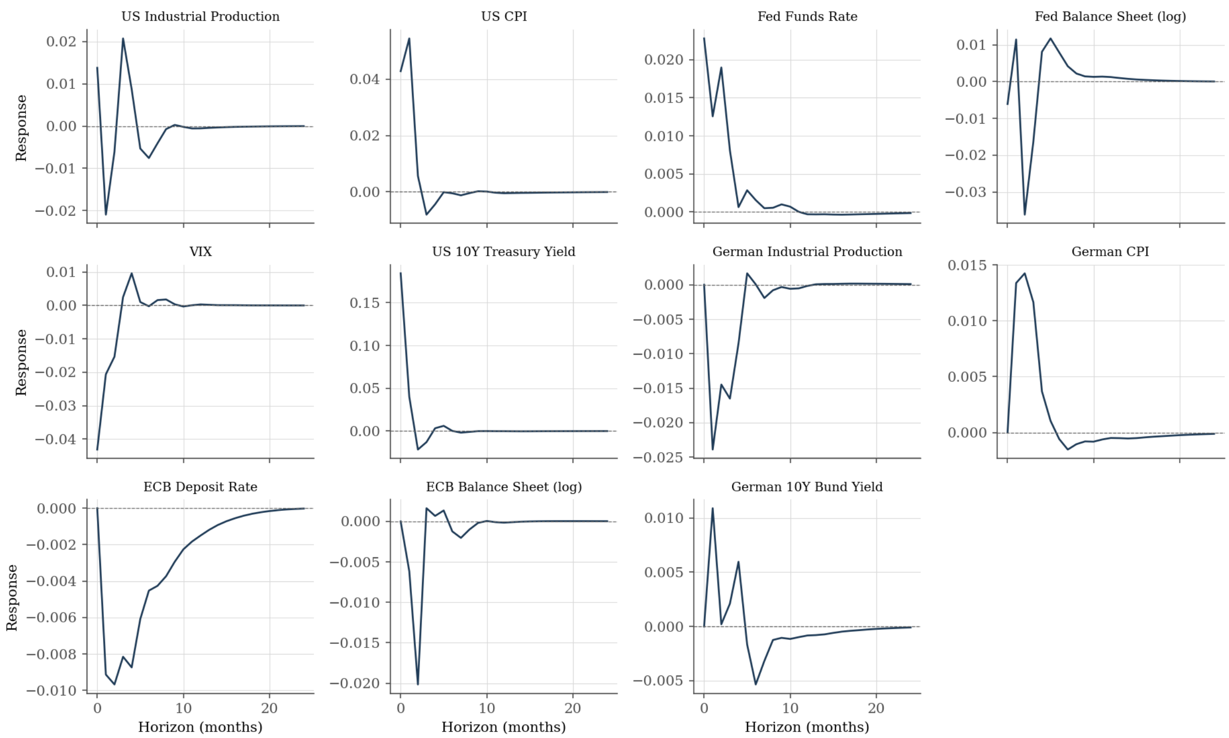

Figure 6: Structural (Cholesky) impulse response of the U.S. 10-year Treasury yield to all eleven shocks in the block-recursive SVAR.

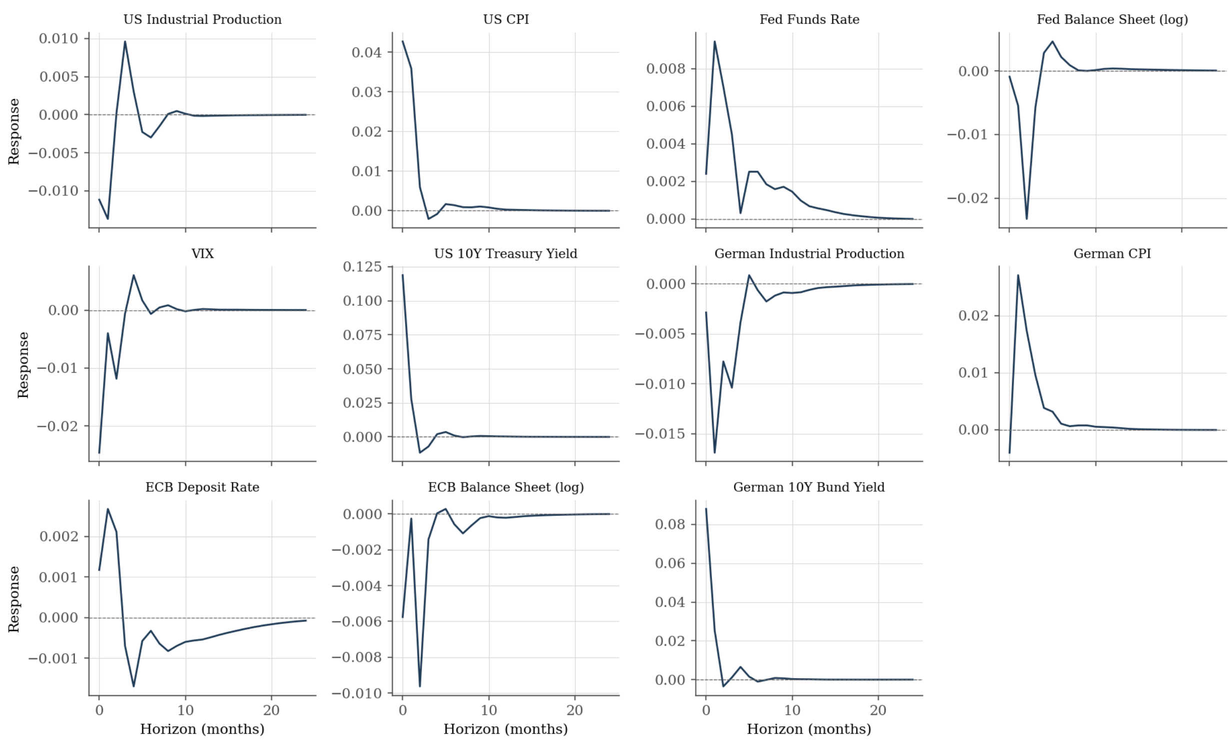

Figure 7: Structural (Cholesky) impulse response of the German 10-year Bund yield to all eleven shocks in the block-recursive SVAR.